RECALL Complex ions AQ-1

Define the term ligand. [1]

RECALL Complex ions AQ-2

Define a co-ordinate (dative covalent) bond. [1]

RECALL Complex ions AQ-3

Define a complex (complex ion). [1]

RECALL Complex ions AQ-4

Define co-ordination number. [1]

RECALL Complex ions AQ-5

Which atom in a water ligand donates the lone pair to the metal ion? [1]

RECALL Complex ions AQ-6

Give the shape and bond angle of $[Cu(H_2O)_6]^{2+}$. [2]

RECALL Complex ions AQ-7

Give the formulas of the four metal-aqua ions in this topic. [1]

RECALL Complex ions AQ-8

What do hexa and aqua mean in “hexaaqua ion”? [1]

RECALL Complex ions AQ-2

1. A shared pair of electrons where **both electrons come from the same atom**

RECALL Complex ions AQ-1

1. An atom, ion or molecule that **donates a pair of electrons** to a central metal ion (forms a co-ordinate bond)

Must say “pair” or “lone pair”. “Donates electrons” alone usually loses the mark.

RECALL Complex ions AQ-4

1. The number of **co-ordinate bonds** to the central metal atom or ion

Not “the number of ligands”: that only works for ligands that form one bond each.

RECALL Complex ions AQ-3

1. A central metal atom or ion **surrounded by ligands** (bonded by co-ordinate bonds)

RECALL Complex ions AQ-6

1. **Octahedral**
2. **90°**

RECALL Complex ions AQ-5

1. The **oxygen** atom

RECALL Complex ions AQ-8

1. Hexa = **six**; aqua = **water** (six water ligands)

RECALL Complex ions AQ-7

1. $[Cu(H_2O)_6]^{2+}$, $[Fe(H_2O)_6]^{2+}$, $[Fe(H_2O)_6]^{3+}$ and $[Al(H_2O)_6]^{3+}$

RECALL Colours AQ-9

Colour of $[Cu(H_2O)_6]^{2+}(aq)$? [1]

Aqueous ions · question

RECALL Colours AQ-10

Colour of $[Fe(H_2O)_6]^{2+}(aq)$? [1]

Aqueous ions · question

RECALL Colours AQ-11

Colour of $[Fe(H_2O)_6]^{3+}(aq)$? [1]

Aqueous ions · question

RECALL Colours AQ-12

Colour of $[Al(H_2O)_6]^{3+}(aq)$? [1]

Aqueous ions · question

RECALL Hydroxide precipitates AQ-13

Observation: NaOH(aq) added to $[Cu(H_2O)_6]^{2+}(aq)$. [1]

Aqueous ions · question

RECALL Hydroxide precipitates AQ-14

Observation: NaOH(aq) added to $[Fe(H_2O)_6]^{2+}(aq)$. [1]

Aqueous ions · question

RECALL Hydroxide precipitates AQ-15

Observation: NaOH(aq) added to $[Fe(H_2O)_6]^{3+}(aq)$. [1]

Aqueous ions · question

RECALL Hydroxide precipitates AQ-16

Observation: NaOH(aq) added to $[Al(H_2O)_6]^{3+}(aq)$. [1]

Aqueous ions · question

RECALL Colours AQ-10

1. **Green** (pale green) solution

RECALL Colours AQ-9

1. **Blue** solution

RECALL Colours AQ-12

1. **Colourless** solution

Write “colourless”, never “clear”. Clear means see-through, which a blue solution also is.

RECALL Colours AQ-11

1. **Purple/violet**, but usually looks **yellow-brown**

It looks yellow-brown because some is hydrolysed to $[Fe(H_2O)_5(OH)]^{2+}$.

RECALL Hydroxide precipitates AQ-14

1. Green solution → **green precipitate**

It then slowly turns brown on standing in air.

RECALL Hydroxide precipitates AQ-13

1. Blue solution → **blue precipitate**

RECALL Hydroxide precipitates AQ-16

1. Colourless solution → **white precipitate**

RECALL Hydroxide precipitates AQ-15

1. Yellow-brown solution → **brown (orange-brown) precipitate**

RECALL Hydroxide precipitates AQ-17

Formulas of the hydroxide precipitates of a 2+ ion and a 3+ ion. [2]

RECALL Hydroxide precipitates AQ-18

What is the overall charge on a metal hydroxide precipitate such as $[Fe(H_2O)_3(OH)_3]$? [1]

RECALL Hydroxide precipitates AQ-19

Why does the green precipitate from Fe^{2+} turn brown on standing? [1]

RECALL Hydroxide precipitates AQ-20

What role does OH^- (or NH_3) play when it forms a hydroxide precipitate? [1]

RECALL Excess ammonia AQ-21

Which of the four hydroxide precipitates dissolves in excess ammonia? [1]

RECALL Excess ammonia AQ-22

Formula and colour of the complex formed with excess NH_3 and Cu^{2+}. [2]

RECALL Excess ammonia AQ-23

Name the type of reaction when NH_3 replaces H_2O ligands. [1]

RECALL Amphoteric aluminium AQ-24

Define amphoteric. [1]

RECALL Hydroxide precipitates AQ-18

1. **Zero**: it is neutral, which is why it is insoluble

RECALL Hydroxide precipitates AQ-17

1. 2+: **$[M(H_2O)_4(OH)_2]$**
2. 3+: **$[M(H_2O)_3(OH)_3]$**

Charge = number of OH. Waters + OH always add to 6. No overall charge.

RECALL Hydroxide precipitates AQ-20

1. It acts as a **Brønsted–Lowry base**: it accepts H^+ from a water ligand

RECALL Hydroxide precipitates AQ-19

1. It is **oxidised** by oxygen in the air from **Fe(II) to Fe(III)**

RECALL Excess ammonia AQ-22

1. **$[Cu(NH_3)_4(H_2O)_2]^{2+}$**
2. **Deep (dark) blue solution**

Order inside the bracket doesn't matter. "Blue" alone is not enough: say deep or dark blue.

RECALL Excess ammonia AQ-21

1. Only the **copper(II)** one, $[Cu(H_2O)_4(OH)_2]$

RECALL Amphoteric aluminium AQ-24

1. Reacts with (dissolves in) **both acids and bases**

RECALL Excess ammonia AQ-23

1. **Ligand substitution** (ligand exchange)

RECALL Amphoteric aluminium AQ-25

Which metal hydroxide in this topic is amphoteric? [1]

RECALL Amphoteric aluminium AQ-26

Formula and appearance of the product when excess NaOH is added to aluminium hydroxide. [2]

RECALL Carbonate AQ-27

Observation: $Na_2CO_3(aq)$ added to $[Cu(H_2O)_6]^{2+}(aq)$. [1]

RECALL Carbonate AQ-28

Observation: $Na_2CO_3(aq)$ added to $[Fe(H_2O)_6]^{2+}(aq)$. [1]

RECALL Carbonate AQ-29

Observations: $Na_2CO_3(aq)$ added to $[Fe(H_2O)_6]^{3+}(aq)$. [2]

RECALL Carbonate AQ-30

Observations: $Na_2CO_3(aq)$ added to $[Al(H_2O)_6]^{3+}(aq)$. [2]

RECALL Carbonate AQ-31

Name the gas given off when carbonate is added to a 3+ aqua ion. [1]

RECALL Carbonate AQ-32

Which ions form a metal carbonate precipitate with Na_2CO_3, and which form a hydroxide? [1]

RECALL Amphoteric aluminium AQ-26

1. **$[Al(OH)_4]^-$** (or $[Al(H_2O)_2(OH)_4]^-$)
2. White precipitate dissolves → **colourless solution**

RECALL Amphoteric aluminium AQ-25

1. **Aluminium hydroxide**, $[Al(H_2O)_3(OH)_3]$

RECALL Carbonate AQ-28

1. **Green precipitate** ($FeCO_3$), no effervescence

RECALL Carbonate AQ-27

1. **Blue-green precipitate** ($CuCO_3$), no effervescence

RECALL Carbonate AQ-30

1. **White precipitate**
2. **Effervescence** (bubbles of gas)

RECALL Carbonate AQ-29

1. **Brown precipitate**
2. **Effervescence** (bubbles of gas)

RECALL Carbonate AQ-32

1. **2+** ions (Cu^{2+}, Fe^{2+}) → MCO_3; **3+** ions (Fe^{3+}, Al^{3+}) → $M(H_2O)_3(OH)_3 + CO_2$

Trick: three fizzes, two doesn't.

RECALL Carbonate AQ-31

1. **Carbon dioxide**, CO_2

RECALL Chloride ions AQ-33

Formula and colour of the complex formed with concentrated HCl and Cu^{2+}. [2]

Aqueous ions · question

RECALL Chloride ions AQ-34

Shape and bond angle of $[CuCl_4]^{2-}$. [2]

Aqueous ions · question

RECALL Acidity AQ-35

Which is more acidic: $[Fe(H_2O)_6]^{2+}$ or $[Fe(H_2O)_6]^{3+}$? [1]

Aqueous ions · question

RECALL Identify & practical AQ-36

Name one reagent that distinguishes $Fe^{2+}(aq)$ from $Fe^{3+}(aq)$, with both results. [1]

Aqueous ions · question

RECALL Chloride ions AQ-34

1. **Tetrahedral**
2. **109.5°**

RECALL Chloride ions AQ-33

1. **$[CuCl_4]^{2-}$**
2. **Yellow** solution (may look green)

RECALL Identify & practical AQ-36

1. NaOH(aq) (or NH_3 or Na_2CO_3): Fe^{2+} **green ppt**; Fe^{3+} **brown ppt**

RECALL Acidity AQ-35

1. **$[Fe(H_2O)_6]^{3+}$** (M^{3+} is always more acidic than M^{2+})